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DIGITAL IMAGE PROCESSING TH.

PROJECT PROPOSAL

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Project Proposal

Person Re-Identification

1. Introduction

Person Re-Identification (ReID) is a computer vision technique used to identify and match a person across different images or video cameras. It is widely used in surveillance and security systems.

This project aims to develop a deep learning-based system that can detect a person in images and videos and recognize whether the same person appears again in another camera or frame.

2. Objectives

- Detect persons from images and video.
- Extract features using **ResNet-50**.
- Match persons using a ReID model.
- Assign consistent IDs across frames or cameras.
- Evaluate system performance.

3. Methodology

Step 1: Person Detection

We will use a pre-trained model like **YOLO** to detect persons in images and video frames.

Step 2: Feature Extraction

Each detected person image will be passed through **ResNet-50** to extract deep features.

Step 3: Person Matching

We will compare extracted features using:

- Cosine similarity or
- Euclidean distance

If similarity is high → same person

If similarity is low → different person

Step 4: Video-Based ReID

For video:

- Detect person in each frame

- Extract features
- Match with stored feature database
- Assign identity

4. Tools & Technologies

- Python
- PyTorch
- ResNet-50
- YOLO (for detection)
- OpenCV

5. Expected Outcome

- System that detects and identifies persons in images and videos
- Ability to track same person across different cameras
- Good matching accuracy

6. Applications

- CCTV surveillance
- Security systems
- Smart city monitoring
- Missing person identification